2025 IEEE/ACIS 29th International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD) | 979-8-3315-1258-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/SNPD65828.2025.11253724

Automated Optical Marker Exam Recognition using Cross-Linear Area Detection from Smartphone Camera

Watcharaphong Yookwan
Faculty of Informatics
Burapha University
Chonburi, Thailand
wyookwan@informatics.buu.ac.th

Athita Onuean
Faculty of Informatics
Burapha University
Chonburi, Thailand
athitha@go.buu.ac.th

Sirapat Tansavetwong
Faculty of Informatics
Burapha University
Chonburi, Thailand
sirapat.ta@go.buu.ac.th

Jirayus Arbking
Faculty of Informatics
Burapha University
Chonburi, Thailand
jirayu.ar@go.buu.ac.th

Sirima Chinnasarn
Faculty of Science
Burapha University
Chonburi, Thailand
sirima@buu.ac.th

Krisana Chinnasarn
Faculty of Informatics
Burapha University
Chonburi, Thailand
krisana@informatics.buu.ac.th

***Abstract*— This paper presents an automated system for evaluating optical marker-based multiple-choice answer sheets using images captured from smartphone cameras. The proposed approach addresses the challenges of document distortion, varying lighting conditions, and diverse sheet templates through a multi-stage pipeline. The system begins with image dewarping using homographic transformation to correct perspective distortions. Key structural components—such as Exam Number, Student ID, and Page Number—are localized using YOLOv8 object detection. The answer region is then identified and cropped for analysis. Template matching is employed to detect landmark points, enabling precise alignment of the answer grid. Optical Mark Recognition (OMR) is performed using a Cross-Linear Area (CLA) algorithm, which calculates the ratio of dark pixels within each marker region to determine marked responses. Detected answers are mapped to corresponding questions and exported in JSON format for downstream use. Experimental results demonstrate that the system is accurate, efficient, and robust to real-world imaging conditions, offering a practical solution for low-cost, mobile-based exam evaluation.**

Keywords—component, formatting, style, styling, insert (*key words****)***

I. Introduction

The automation of examination grading processes has become increasingly vital in modern educational and certification systems, aiming to reduce human error, minimize processing time, and enhance overall efficiency. Traditional Optical Mark Recognition (OMR) technologies, while effective, typically depend on specialized scanning equipment and standardized answer sheets, leading to significant infrastructural costs and limited adaptability in diverse operational environments [1]. In recent years, the proliferation of high-resolution smartphone cameras has opened new avenues for developing low-cost, portable OMR solutions accessible to a broader range of users [2].

This paper presents an automated optical marker exam evaluation system utilizing a novel Cross-Linear Area (CLA) algorithm, designed specifically for smartphone-based image acquisition. In contrast to conventional OMR techniques, which primarily rely on pixel intensity analysis, thresholding, or simple contour detection, the proposed CLA algorithm evaluates the distribution of filled areas across predefined cross-linear scan paths. This method enhances detection robustness against common challenges such as variable illumination, non-uniform shading, form distortions, and imprecise user markings, which frequently impair the reliability of traditional approaches.

The proposed system enables users to capture examination sheets using standard smartphone cameras without requiring extensive alignment procedures or highly controlled lighting conditions. Subsequent processing steps involve form normalization, marker localization, and response recognition based on the CLA algorithm, culminating in automatic scoring. Experimental validation across various smartphone models, lighting scenarios, and form designs demonstrates the system's effectiveness and resilience, achieving high accuracy comparable to that of conventional OMR systems while offering greater flexibility and significantly reduced operational constraints.

II. Problem Statements

Despite the widespread adoption of Optical Mark Recognition (OMR) systems for automated exam grading, traditional OMR solutions are inherently constrained by several critical limitations [3]. Firstly, they require specialized hardware such as high-precision scanners and uniformly printed answer sheets, resulting in high deployment and operational costs. Secondly, conventional OMR techniques exhibit sensitivity to variations in environmental conditions, including uneven lighting, paper distortions, and incomplete or ambiguous markings, which can significantly degrade recognition accuracy [4]. These constraints make traditional systems less accessible, particularly in resource-limited settings such as remote educational institutions or informal testing environments [5].

The advent of smartphone technology, with its ubiquitous availability and continual advancements in camera quality, presents an opportunity to democratize automated exam evaluation by eliminating the dependency on specialized equipment [6]. However, leveraging smartphone-captured images introduces new technical challenges, including variable lighting, perspective distortions, occlusions, and inconsistent answer sheet formats. Traditional threshold-

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based or contour-based detection methods often fail to maintain acceptable accuracy under such uncontrolled conditions.

Accordingly, there is a pressing need for a robust, low-cost, and accessible OMR solution capable of operating effectively with images captured under diverse real-world conditions using standard smartphone cameras. To address these challenges, this paper proposes a novel system based on the Cross-Linear Area (CLA) algorithm, which is specifically designed to enhance the reliability of mark detection by analyzing structural area distributions rather than relying solely on pixel intensity or contour shapes. This approach aims to deliver a practical, accurate, and scalable solution for automated exam evaluation without the necessity for specialized hardware or strict environmental control.

III. Background

Before Optical Mark Recognition (OMR) is a widely used technique for the automated reading of marked data from paper forms, particularly in standardized testing and survey collection. Traditional OMR systems typically rely on high-resolution scanners and fixed-form layouts, which require precise printing and alignment. These systems, while accurate, are costly, infrastructure-dependent, and lack portability—posing limitations in decentralized or resource-limited educational settings [7].

With the increasing availability and performance of smartphone cameras, mobile-based OMR systems have gained attention as cost-effective alternatives. However, such systems must overcome several challenges, including variations in lighting, perspective distortion, and inconsistent user handling [8]. To address these challenges, recent work has incorporated computer vision and deep learning methods to enhance document capture robustness and response recognition.

Object detection models such as YOLO (You Only Look Once) have shown strong performance in localizing objects in real-time applications. YOLOv8, the latest iteration, offers improved accuracy and speed [9], making it suitable for mobile deployments. Additionally, template matching remains a reliable method for detecting fixed visual features, such as alignment markers, when geometric consistency is required.

This work builds upon these advancements by integrating image dewarping, YOLOv8-based field localization, template-based landmark detection, and a custom Cross-Linear Area (CLA) algorithm to enable reliable, end-to-end OMR processing directly from smartphone-captured images.

IV. Methodology

The proposed Automated Optical Marker Exam Evaluation System is designed to operate end-to-end from image acquisition to structured output generation, leveraging standard smartphone cameras and the novel Cross-Linear Area (CLA) algorithm for robust mark detection. The complete methodology consists of several sequential stages, as detailed in Fig 1.

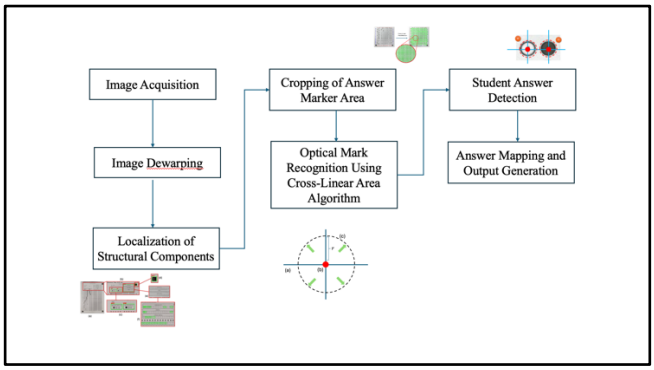


Fig. 1. Overall of Proposed Method

A. Image Acquisition

The examination answer sheet is captured using a smartphone camera. To enhance robustness against environmental variability, users are guided to position the sheet within a visible frame, but strict alignment is not mandatory. Captured images are stored at high resolution to preserve detail necessary for subsequent processing.

B. Image Dewarping

Smartphone-captured images often exhibit perspective distortions that must be corrected to ensure accurate optical mark recognition. The system performs image dewarping through homography-based transformation [10]. Initially, edges are detected using the Canny algorithm, followed by contour detection to identify the sheet boundary, assumed as the largest quadrilateral. The four corner points — Top-Left (TL), Top-Right (TR), Bottom-Right (BR), and Bottom-Left (BL) — are extracted and ordered. A homograph matrix $\mathbf{H} \in \mathbb{R}^{3\times3}$ is then computed to map the distorted quadrilateral to a rectangular reference frame, satisfying Equation (1).

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{1}$$

where (x, y) are the original coordinates and (x', y') are the corrected coordinates. The matrix H is estimated using the Direct Linear Transformation (DLT) method from four point correspondences. Finally, the image is warped using the inverse homography $\mathbf{H}^{-1}$ to obtain a rectified, top-down view suitable for further processing. Fig. 2 demonstrated an input (a) and the dewarped image (b).

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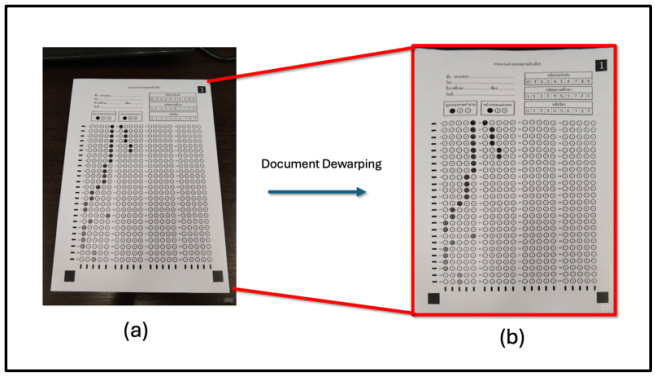


Fig. 2. Document Dewarpping, input image (a), output image (b)

C. Localization of Structural Components

After dewarping, the system uses YOLOv8 object detection to localize key structural components on the answer sheet, including Exam No., Page No., School ID, Subject ID, Student ID, and Template No. YOLOv8 is trained on annotated answer sheet images, with each field labeled as a unique class. During inference, the model detects bounding boxes $b_i = (x, y, w, h, c)$. for each component. Non-Maximum Suppression (NMS) is applied to filter overlapping detections. The output in Fig. 3 (a) – (f), consequently, includes the bounding box and associated class, which is used to extract and interpret metadata content.

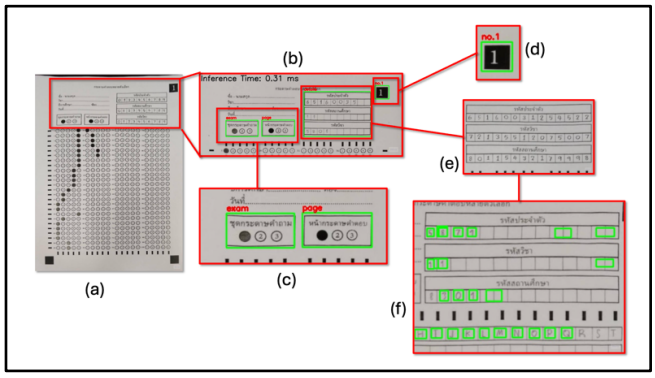


Fig. 3. Components identification

D. Cropping of Answer Marker Area

After component localization, YOLOv8 is used to detect the region containing all candidate answer markers, labeled as a single *answer_area* class. The model is trained on annotated sheets with bounding boxes enclosing the entire answer grid. During inference, YOLOv8 outputs a bounding box that is used to crop the answer region from the dewarped image. This cropped region is then passed to the landmark detection and OMR module for further analysis.

E. Cropping of Answer Marker Area

To accurately identify the position and orientation of answer markers within the cropped answer region, the system employs template matching for landmark detection. These landmarks—typically distinct shapes. Cross markers—serve as positional anchors for aligning the grid of response markers.

a) Landmark Templates

Template images are created in advance from representative answer sheets, capturing the expected appearance of landmarks (e.g., corners or bounding marks of the answer grid). Each $T \in \mathbb{R}^{w \times h}$ is a grayscale patch corresponding to a known landmark.

b) Matching Process

The cropped answer area $I \in \mathbb{R}^{H \times W}$ is scanned using normalized cross-correlation to compute the matching score at each location using Equation (2).

$$R(x,y) = \frac{\sum_{i,j} \left[I(x+i, y+j) - \bar{I}_{x,y}\right][T(i,j) - \bar{T}]}{\sqrt{\sum_{i,j} \left[I(x+i, y+j) - \bar{I}_{x,y}\right]^2 \sum_{i,j} [T(i,j) - \bar{T}]^2}} \quad (2)$$

Where:

$R(x, y)$ is the correlation score at location (x, y)

$\bar{I}_{x,y}$ is the mean of the image patch under the template,

$\bar{T}$ is the mean intensity of the template.

The final landmark coordinates are used to infer the grid geometry (rows, columns, spacing), allowing precise alignment of the cross-linear areas over the answer bubbles as shown in Fig. 4.

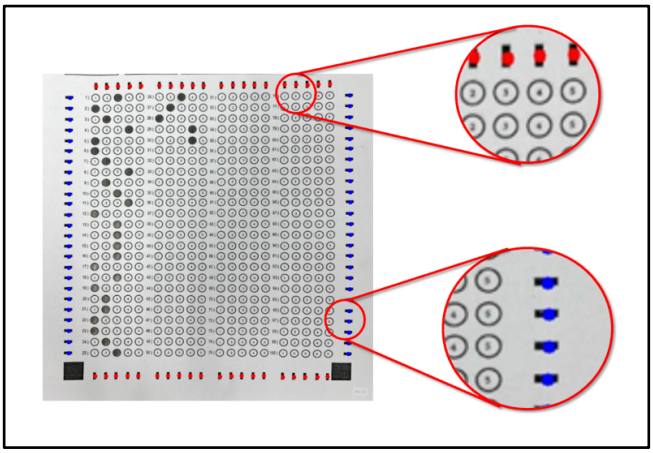


Fig. 4. Landmark Detection

F. Optical Mark Recognition Using Cross-Linear Area Algorithm

The core of the recognition process utilizes the proposed Cross-Linear Area (CLA) algorithm. Each marker region is systematically scanned along multiple linear cross-sections, and the density of dark pixels along these paths is analyzed. The CLA approach mitigates errors caused by uneven shading and partial markings, offering superior robustness compared to conventional global thresholding. After landmarks were identified, the line between the opposite landmark were created as demonstrated in Fig. 5.

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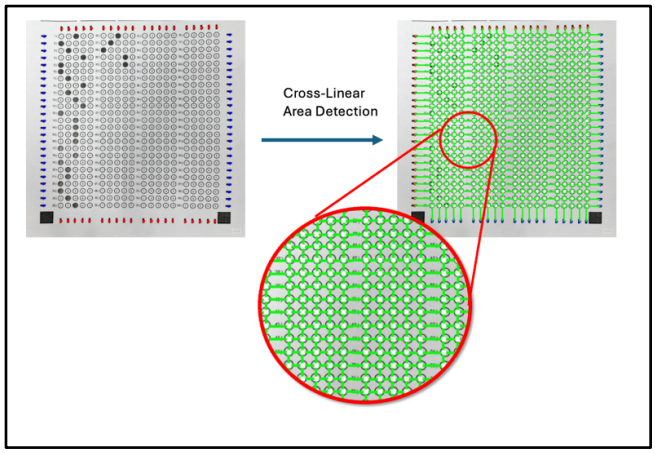


Fig. 5. Optical Mark Recognition Using Cross-Linear Area

By using the cross line from landmark, Fig. 6 shows the interested area (c) to determination is expanded by r circularly from centroid (b).

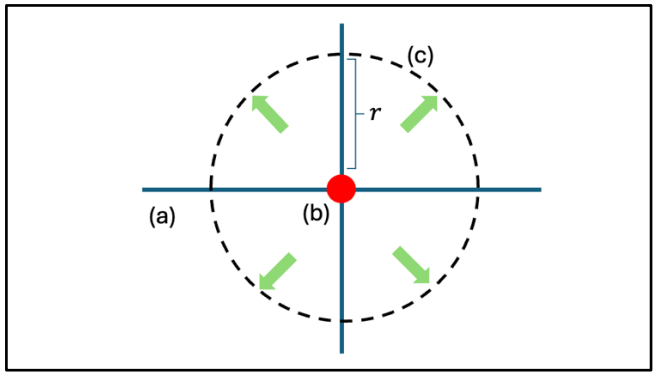


Fig. 6. Centroid Area Expansion

G. Student Answer Detection

For each marker, the ratio of dark pixels detected within the cross-linear paths is calculated. Fig. 7 demonstrated if the ratio exceeds a predefined threshold, the marker is classified as "filled" (a) Otherwise, it is considered "unfilled" (b). This binary classification is performed independently for each answer option.

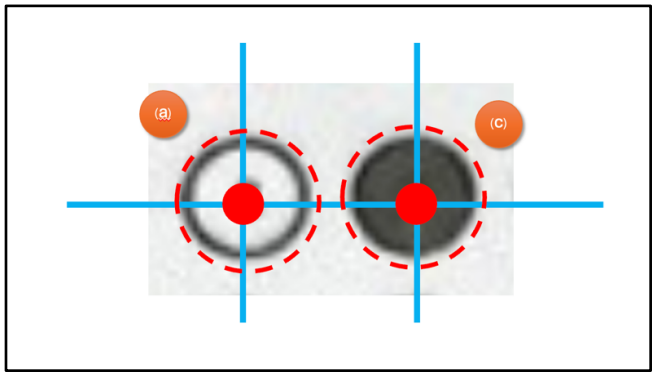


Fig. 7. Student Answer Detection

The algorithm of Cross-linear and Answer Detection can be described as *Algorithm I*

Algorithm: Cross-Linear Area OMR
Input: List of Landmarks *Landmark*
BEGIN:
1: **FOREACH** *Landmark*
2: **COMPUTE** opposite $Landmark_i \rightarrow Landmark_j$
3: **COMPUTE** Linear line l_{ij} FROM $Landmark_i$ *AND* $Landmark_j$
4: **COMPUTE** Centroids for each crossed line $Centroid_{ij}$
5: **EXPAND** Area around each $Centroid_{ij}$ by r
6: **COMPUTE** Dark Pixel ratio In the expaned area
7: **MAP** Answer Ans_i as JSON file
END:
Output: Set of answers in JSON format

H. Answer Mapping and Output Generation

Detected answers are mapped to their corresponding questions based on the predefined template layout. The final output, including metadata (Exam No., Student ID, etc.) and student responses, is structured in JSON format for easy integration with external systems such as grading servers, educational databases, or learning management systems.

V. Experimental Result

To validate the effectiveness of the proposed system, experiments were conducted on a dataset of 500 answer sheets captured using various smartphone models under diverse real-world conditions. The dataset included five distinct answer sheet templates with variations in lighting, perspective distortion, and paper alignment to simulate practical usage scenarios. The performance can be detailed in the Table below.

TABLE I PERFOMANCE DETAILS

Component	Accuracy (%)	Precision (%)	Recall (%)
Structural Field Detection (YOLOv8)	98.4	97.9	98.7
Answer Area Detection	99.2	99.0	99.5
Landmark Detection (Template Matching)	97.5	—	—
Answer Mark Recognition (CLA OMR)	96.8	95.4	97.9

The performance of the system was evaluated based on several key criteria, including structural field detection accuracy, answer area localization, landmark detection, and Optical Mark Recognition (OMR) accuracy using the Cross-Linear Area (CLA) algorithm. The YOLOv8 model achieved a structural field detection accuracy of 98.4%, with a precision of 97.9% and recall of 98.7%. The answer area was correctly localized in 99.2% of cases, while landmark detection using template matching achieved an accuracy of 97.5%. The CLA-based OMR module yielded an answer recognition accuracy of 96.8%, with a precision of 95.4% and recall of 97.9%. The average end-to-end processing time, from image input to structured JSON output, was measured at 1.35 seconds per sheet on a mid-range smartphone (Snapdragon 778G, 6GB RAM). The system demonstrated

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robustness against skewed inputs up to 20 degrees and performed reliably under moderate shadows and varied lighting conditions. Most errors were attributable to inconsistently marked bubbles, severe lighting gradients, or sheets with physical damage such as folds or ink smudges. These issues were partially addressed by incorporating adaptive thresholding in the CLA algorithm and additional pre-filtering for low-contrast regions. Overall, the results indicate that the system is both accurate and efficient, making it well-suited for mobile-based, scalable exam evaluation in real-world environments.

VI. Conclusion and Discussion

This paper presented a smartphone-based automated system for evaluating multiple-choice exam answer sheets using a combination of modern computer vision techniques, including YOLOv8 object detection, template matching, and a Cross-Linear Area (CLA) algorithm for Optical Mark Recognition (OMR). The proposed system effectively addresses common challenges in mobile-based document processing, such as perspective distortion, template variability, and inconsistent lighting. Experimental results on a diverse dataset of answer sheets demonstrated high accuracy in field detection and answer recognition, as well as fast processing speeds suitable for real-time use on mid-range mobile devices. The modular architecture, comprising dewarping, component localization, answer region detection, landmark alignment, and OMR, enables the system to be easily adapted to different templates and deployment contexts. Despite the strong performance, the system has some limitations. Recognition accuracy may be affected by heavily smudged or partially filled markers, and performance can degrade under extreme lighting conditions or severe image blur. Future work will focus on integrating adaptive illumination correction, real-time feedback during capture, and end-to-end deep learning approaches for bubble detection to further improve robustness and usability.

In conclusion, the system offers a low-cost, portable, and scalable solution for automated exam evaluation, particularly valuable in resource-constrained educational environments where traditional OMR scanners are impractical or unavailable.

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